

A REVIEW PHARMACEUTICAL QUALITY AUDITS

¹Devendra S. Mahale*, ²Apurva S. Patel, ³Jagruti R. Girase,

⁴Kiran P. Patil, ⁵Dr. Sunila A. Patil

^{1,2,3,4}Student, P.S.G.V.P. Mandal's College of Pharmacy, Shahada (Maharashtra), India

⁵Assistant Professor, P.S.G.V.P. Mandal's College of Pharmacy, Shahada (Maharashtra), India

Corresponding Author: -

Email Id :- devendramahale2306@gmail.com

ABSTRACT

These days, auditing is an essential role in a pharmaceutical company. A quality audit aims to improve a quality system by reviewing and evaluating all or a portion of it. It's a tool for reviewing pharmaceutical programs and making sure that the protocols and payment systems meet legal and regulatory standards. A team chosen by management to carry out this task or external, independent experts typically perform quality audits. It is possible to expand these audits to include contractors and suppliers. An audit will evaluate the benefits to the company of strengthening processes and creating a better system by analyzing the strengths and weaknesses of quality assurance and quality assurance processes. This article covers a number of topics related to quality auditing in the pharmaceutical sector, such as its goals, planning, significance, advantages, and planning, as well as potential flaws that may arise. Using the Google Scholar search engine and the following keywords, this review provides a well-organized summary of the different guidelines that have been made public up to this point.

Key Words: - Audit, International organization for standardization (ISO), Internal audit, External audits, Regulatory audits.

INTRODUCTION

Reviews are carried out to determine the accuracy and consistency of the data as well as to evaluate a system's internal control. It gives management insight into how effectively the business maintains quality control over its operations and output [1]. To put it simply, an audit is the examination of a system or process to make sure it satisfies the needs of the intended use [2]. "Systematic, independent and documented process for obtaining audit evidence and evaluating them objectively to determine the degree to which the verification criteria are met" is how the International Organization for Standardization (ISO) describes audits [3]. Pharmacies ought to regard the audit as a quality control tool rather than as an intrusive and maybe dangerous

examination. The audit's findings and the ensuing corrective measures reassure all parties concerned that a program operates in compliance with accepted standards of practice [4].

Over the past two years, a significant advancement in operations management has been the application of quality audits to evaluate the efficacy of quality management systems in manufacturing and service companies. Quality is the sum of all the characteristics that make up a good or service and show that it can meet a demand, whether explicit or implicit (quality refers to a degree or level of excellence). The Latin origin of the term "audit" is "the act of hearing." These audits offer an impartial evaluation of the

organization's operating systems' efficacy and adherence to predetermined criteria, including the ISO 9001-2000 quality standard [5].

Audits are defined by the International Organization for Standardization (ISO) as

a systematic, unbiased, and recorded procedure. It entails obtaining audit evidence and impartially assessing it to ascertain the degree to which the verification requirements are satisfied. [6,7]

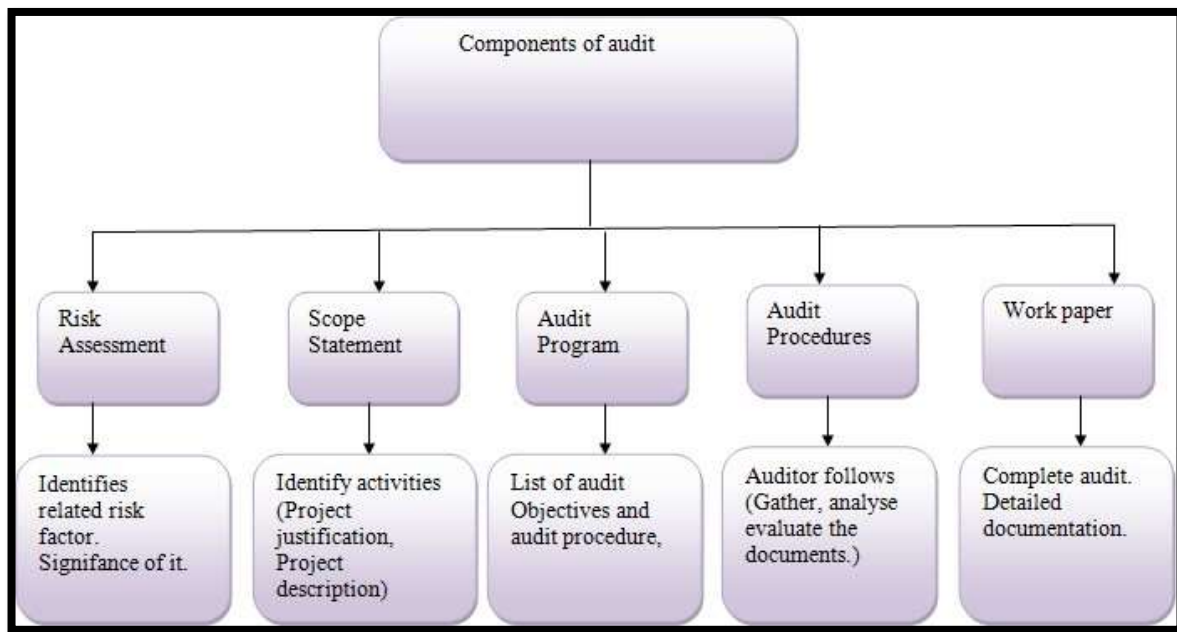


Fig.1: Components of audit.

Principles of auditing: -

The audit is distinguished by its reliance on certain guiding concepts. Because they offer guidance on actions an organization may take to enhance performance, these principles should contribute to the establishment of audit as a trustworthy and efficient instrument to support management policies and controls. Respecting these guidelines is essential to producing adequate and pertinent audit conclusions as well as enabling auditors to operate independently of one another to arrive at comparable results under comparable conditions.

A straightforward definition of an audit would be the examination of a system or process to make sure it satisfies the needs of its intended purpose. The audits are described as "Systematic, independent, and documented process for obtaining audit

evidence and evaluating them objectively to determine the degree to which the verification criteria are met" by the International Organization for Standardization (ISO).

The modern definition of auditing is conducting an official examination or testing against predetermined criteria. "A precise and free assessment to decide if quality exercises and results consent to arranged game plans and whether these plans are carried out really and are reasonable for accomplishing goals" is how the International Standard ISO 10011-1 (1990) defines the term "quality audit" [8].

Definition of internal audit: - ^[9]

Internal audit is described as "an independent appraisal activity established within an organization as a service to it"

by the chartered institute of management accountants, UK (CIMA). It is a management tool that assesses the efficacy of every aspect of an organization's management and operations. It works by looking at and assessing the suitability and efficacy of other controls.

Similar definitions of internal auditing may be found in the Institute of Internal Auditors' (IIA) definition, which states that internal auditing is an impartial, independent assurance and consulting activity intended to enhance an organization's operations. By providing a methodical, disciplined approach to assess and enhance the efficacy of risk management, control, and governance procedures, it aids a business in achieving its goals.

Two distinct roles of the internal audit activity are stated in these definitions, namely.

Internal control: A procedure that is carried out by the company's staff members and the information technology systems that help the business achieve its goals.

Management tool: These keep an eye on and assess how well a business manages its risks and its operating procedures [10].

Goals of an audit: -

This intricate process's straightforward objective is to assess current operations and paperwork to see if they adhere to the set standards. The purpose of an audit is to assess the advantages and disadvantages of the company's quality control and quality assurance procedures. Based on the audit's findings, we will be able to enhance the processes and construct a more robust system. A pharmaceutical company's products all have qualities that need to be determined or verified through laboratory testing. In the pharmaceutical sector, quality assurance and control are essential

procedures that function as a control and balance system.

The audit process itself should have little trouble accomplishing its goal with the right planning and preparation. Building brand identity and avoiding the negative consequences of non-compliance, including as fines, bad press, and legal actions, can be achieved by efficient audits and proper adherence to standards [11].

Objectives of audit: -

- i. Establish effectiveness to assemble the quality requirements and quality objectives.
- ii. Assembles the GMP and regulatory requirement.
- iii. Providing an opportunity to audit team to improve quality management system^[12].
- iv. Permitting list of audited registered companies in a register.
- v. Determining conformity or non-conformity to assemble specified quality system.
- vi. Modifying potential of sellers.
- vii. Assembles conformity of implementation of documents^[13,14].
- viii. Evaluating conformity of requirements to ISO 9001
- ix. Evaluating conformity of documentation to ISO 9001
- x. Judging conformity of implementation to documentation
- xi. Determining effectiveness in meeting requirements and objectives
- xii. Meeting any prescribed or regulatory requirements for auditing
- xiii. Providing an opportunity to improve the quality management system
- xiv. Permitting registration and inclusion in a list of registered companies
- xv. Qualifying potential suppliers^[15].

Benefits of auditing: -

Audits should be viewed as a management tool to evaluate the organization's internal quality management system, even if they

typically have little bearing on regulatory inspections. Both external and internal audits can assist in achieving this objective. An efficient audit system's main advantages can be summed up as follows: Taking charge of a quality control system identifying unfavorable trends or events in order to identify weak points in advance preventing quality failures through the examination of quality data.

Notifying Senior Management on the state of the operations and/or facilities Standardizing audits will maximize output, raise audit quality standards (and hence, product and service quality), and ultimately create a feedback loop for continuous development.

The audit will comprehend that the purpose of the audit is to enhance the performance of the organization, not to monitor and critique his work. As a result, the audits will be accepted more widely. He will view audits as an opportunity to learn more and broaden his understanding of quality-related topics.

The audit will be more accepting and save time if quality, safety, and environmental audits are combined. This will result in a large reduction in the number of audits. Pooling audits, such as through Shared Third Party Audits, can yield additional benefits.

A superior audit system will be established across the industry, raising the degree of compliance. These efforts will lead to increased mutual confidence and better relationships between the partners [16]. Running a system for quality control identifying unacceptable tendencies or circumstances in order to anticipate weak points. Preventing quality lapses through the examination of quality data.

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Eliminate obstacles to ongoing improvement.

Convert reactive to proactive actions.

Collect information to help quality systems get better.

Types of audits: -

Three primary categories comprise the quality audit system:

Internal Audits

External Audits

Regulatory Audits

To confirm a quality management system's efficacy, quality audits are carried out [17].

Internal audit

First-Party Audit and self-audit are other names for this kind of audit. Both the auditing and the audited parties are affiliated with the same organization [18]. One of the professional activities of internal auditing is giving advice to organizations on how to better accomplish their objectives. The internal audit entails

analyzing organizational issues or company processes using a methodical approach and making recommendations for improvements. The following is a summary of internal audits' primary goals: To support the system of internal controls evaluation of the functioning of organizational policies. Check for mistakes and frauds to ensure they are genuine and accurate.

Error and defect detection and prevention. Protecting the resources the suitability of accounting regulations. Facilitates the efficient operation of the internal check system. Internal auditing in a pharmaceutical factory primarily involves two checks that must be made: Tasks completed by various departments and the records these departments keep. A detailed document list and department-specific questionnaire must be created for this reason [19].

External audits

Second-Party Audit is another name for this kind of audit. It alludes to a consumer auditing a contractor or supplier [18]. Even so, this control is not subject to any stringent legal restrictions. It is usually a good idea to assess the level of skill of the contractors we use to manufacture our products, analyze them, or do any other GMP-compliant task. Additionally, conducting these audits has significant business benefits:

Gain understanding of and trust in the cooperation agreement. Guarantees that specifications are recognized and fulfilled permitting the decrease of specific tasks can lower the chance of failure (and, consequently, its costs)—for example, internal quality control (QC) testing of starting materials.

Numerous suppliers to the pharmaceutical business hold ISO 9001 or ISO 9002 certifications, and their certifying organization conducts routine audits of them. Businesses that contract to

manufacture or package pharmaceuticals will have to get licenses and be subject to regulatory audits [19].

Advantages are as follow:

Increase understanding and confidence in the collaboration contract. Makes certain that requirements are recognized and accepted. Reduce the possibility of failing.

Regulatory audits

Third-Party Audit is another name for this kind of audit. Customers and suppliers don't carry out these kinds of audits. A third-party audit is carried out by an independent organization or regulatory body for registration, certification, or compliance-related reasons [18]. International regulatory agencies such as. These inspections are conducted by the Medicines and Healthcare Products Regulatory Agency (MHRA), UK, US Food and Drug Administration (USFDA), Therapeutic Goods Administration (TGA), Australia, Medicines Control Council (MCC), South Africa, etc. The audit will be carried out by a team that includes a multidisciplinary company staff and audit inspectors.

Every department in the organization, including manufacturing, quality control, warehouse, maintenance, administration/personnel, and marketing/sales, needs to be represented. Since manufacturers must always follow GMPs, these audits can be carried out without prior notice (MHRA presently conducts about 10% of its inspections in the UK in this manner). Companies may also be audited by regulatory agencies in other nations where goods are sold (for example, the FDA inspects manufacturers in Europe).

Regulatory inspectors are all very skilled, knowledgeable, and kind. Every MHRA inspector is professionally certified and has worked in a production operation for at

least five years. Their names will be on the lists of those who can act as lead auditors and qualified individuals.

A manufacturing license or an import/export license may be restricted or revoked for not approving a regulatory audit. (Recently, financial organizations that failed to comply with GMPs and respond appropriately to audit results have been subject to "punitive consensus decrees" from the FDA). As a result, it is crucial that businesses have clear procedures for handling audits, and employees need to receive sufficient training on auditing procedures. Internal audits might offer beneficial chances for practice [19].

These audits are carried out by regulatory organizations like the USFDA and MCA and are done in secret. A official report will be given following the regulatory audit.

For MCA: Verbal feedback report is given at exit meeting.

For USFDA: Provides feedback in form no 483 at exit meeting.

The audit plan

To bring audit risk down to a manageable level, the auditor should create an audit plan for the audit. The audit plan, which goes into greater detail than the audit strategy as a whole, specifies the kind, time, and scope of the audit procedures that team members must carry out in order to gather enough relevant audit evidence to lower the audit risk to a manageable level. The audit plan's documentation acts as a record of the appropriate planning and execution of the audit processes, which may be examined and approved before carrying out more audit operations [20].

Planning objectives

"The goal of the evaluator is to design the review so it will be acted in a successful

way." Audits have the potential to be costly, hazardous, and complicated procedures. Even when companies have standardized processes and internal manuals, engagements must be prepared to guarantee that the auditor: Gives crucial areas of the audit the proper focus; promptly recognizes and addresses possible issues;

Arranges and oversees the audit to ensure that it is carried out effectively and efficiently;

Chooses team members who possess the necessary qualities and competencies;

Oversees, manages, and evaluates the team's work;

Efficiently manages the work of other people, including internal audit and experts.

Auditing procedure

The audit process consists of ten steps in total:

Notification: Notification is the first step in the audit process. The notification procedure notifies the party that will be audited of the process's date and time. The notification will also include a list of the papers the order requests to inspect in order to comprehend the company's organizational structure.

Planning: Planning is the process by which an auditor identifies important areas of concern and risk prior to conducting an audit.

Opening meeting: Meeting between the auditing team and the auditing target's administrative and senior management. The auditors will outline the procedure they plan to follow. The management will outline to them the areas of concern and the employees' schedules that need to be consulted.

Fieldwork: Following the conference, fieldwork is conducted in order to refine the final audit plans. Workers are informed about the audit, schedules for the audit staff's activities are created, and an initial

inquiry is launched after learning about business procedures, key staff interviews, a sample test of current business practices, a review of the law, and a rationality check of internal rules and procedures.

Communication: In order to obtain access to records, explain procedures, and get clarification on processes, the audit team should communicate with the corporate auditor on a regular basis.

Draft audit: The audit is completed, and then the draft audit is prepared. The draft audit includes a list of issues, a distribution list for parties to get preliminary results, and an explanation of what was done and discovered.

Management response: The management team reviews, edits, and suggests changes to the document, looks at areas of concern, and fixes errors. The report is sent to management for the seventh phase, the management reaction, once any last adjustments have been made. In response to the report, management is asked to indicate whether or not they agree with the issues raised, their plan for resolving the issues raised, and when they anticipate that all of the issues will have been resolved.

Final meeting: The purpose of the last meeting is to settle any outstanding issues, go over the management's response, and talk about the audit's scope.

Report distribution: Report dissemination, the ninth phase, involves sending the completed audit report to the relevant authorities both inside and outside the audit region.

Feedback: The audited organization implements the suggested adjustments as part of the last step, known as the audit feedback, and the auditors evaluate and assess the effectiveness, observance, and quality of the implemented modifications. This keeps going until every problem is accepted and the subsequent audit cycle starts [21].

Auditor's responsibility

The following are the duties of the auditor: Help choose the squad and keep the team informed.

Having to organize and oversee every stage of the audit.

With the audit, represent the audit team.

Regulate disputes and handle challenging circumstances.

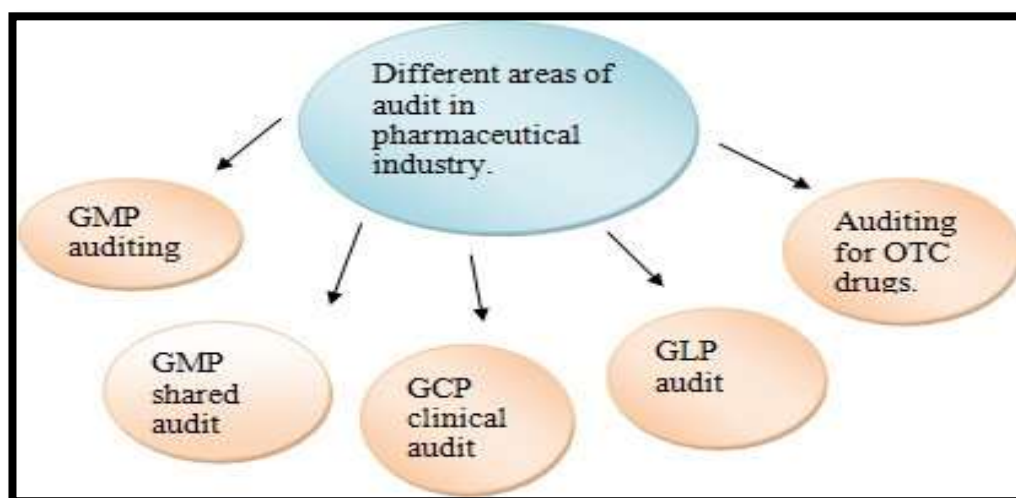
Oversee and lead all discussions with the team and the audit.

Make choices on the quality system and audit issues.

Report the audit's findings as soon as possible.

Describe the primary issues that were resolved.

Report serious inconsistencies right away
possesses strong communication abilities.



Different Types of Audit ^[21]

Conclusion

According to a quality systems approach, audits must be carried out at predetermined intervals to assess how well the quality system is implemented and maintained and to ascertain whether the processes and products satisfy defined requirements. An examination carried out by an experienced and well-prepared auditor can be quite helpful in identifying areas that actually need improvement.

An audit is a comparison of what is written down and what is in place; it should not be interpreted as an interrogation with the audit as a permanent loser. Auditing has no inherent purpose. In the pharmaceutical industry, auditing has two distinct purposes: business requirements and regulatory compliance. Employees and management start to view auditors as opportunities for improvement rather than as police officers when they start to regard audits as such.

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